Type of solutions w/ boundary value problems

Given the general solution, $y = Ae^{x}\cos(2x) + Be^{x}\sin(2x)$

Example 1: [Unique particular soln]

Subject to $y(0) = 0$, $y(1) = 3$

$\therefore$ When $x = 0$, $y = 0$

$0 = A \implies A = 0$

When $x = 1$, $y = 3$

$3 = 0 + Be^{1}\sin(2)$

$B = \dfrac{3}{e\sin(2)}$

$\therefore$ Particular solution: $y = \dfrac{3}{e\sin(2)}\left[e^{x}\sin(2x)\right]$

Example 2: [Infinitely many particular soln]

Subject to $y(0) = 0$, $y(\pi) = 0$

$\therefore$ $y(0) = 0 = A$

$\implies A = 0$

$y(\pi) = 0 = 0 + Be^{\pi}\sin(2\pi)$

$\implies 0 = 0$ [For any B]

$\therefore$ Particular soln: $y = Be^{x}\sin(2x)$ for any B

Example 3: [No solution]

Subject to $y(0) = 0$, $y(\pi) = 1$

$\therefore$ $y(0) = 0 = A$

$\implies A = 0$

$y(\pi) = 1 = Be^{\pi}\sin(2\pi) = 0$

$\implies 1 = 0$ [Contradiction]

$\therefore$ No solution.

Eigenvalue ODE problem

Example 1 :

Consider $\frac{d^2y}{dx^2} + \lambda y = 0$, subject to $y(0) = 0$, $y(1) = 0$

For which values of λ are there infinatelly many solutions ?

Rearranging the ODE , we get

$$\frac{d^2}{dx^2}(y) = (-\lambda) y$$

[Notice this is similar w/ the linear algebra e.vec form of $Ax = \lambda x$]

Why is this called an eigenvalue problem ?

special inputs [e.vec / e.func.]

$$Ax = \lambda x$$

Operator [Matrix / $\frac{d^2}{dx^2}$]

scalar [e.val]

[General Form]

Linear algebra : When transformation matrix A acts on $\vec{x}$, it scales $\vec{x}$ (e.vector) w/out changing its direc.

ODE : When $\frac{d^2}{dx^2}$ acts on y , it scales $y(x)$ (e.function) w/out changing its shape

$\boxed{\text{When } \lambda = 0}$

$\frac{d^2y}{dx^2} = 0$

$\frac{dy}{dx} = A$

$y = Ax + B$

Using boundary conditions,

$y(0) = B = 0$

$y(1) = A + 0 = 0$

Particular solution

$y(x) = 0$

$\therefore$ The soln literally means $\frac{d^2}{dx^2}(0) = 0(0) \Longrightarrow 0 = 0$.

Although this is true , it doesn't give any useful insights .

This is the trivial soln we do not want .

Just like how we don't want the trivial solution when solving linear algebra eigenvector.

When $\lambda > 0$

$$\frac{d^2y}{dx^2} + \lambda y = 0$$

Auxilary Eqn

$r^2 + \lambda = 0$

$r = \pm\sqrt{-\lambda}$

$= \pm\sqrt{\lambda}\, i$

$= \pm w i$, where $w = \sqrt{\lambda}$

General solution

$y(x) = A\cos(wx) + B\sin(wx)$

Using boundary condition,

$y(0) = A + 0 = 0 \implies A = 0$

$y(1) = 0 + B\sin(w) = 0$

$\implies B = 0$ or $\sin(w) = 0$

[Reject. Gives trivial soln] same like $\lambda = 0$

$w = n\pi$, where $n \in \mathbb{Z}^+$

Particular solution

$y = B\sin[(n\pi)x]$, for $w = n\pi \implies \sqrt{\lambda} = n\pi$

$\lambda = (n\pi)^2$ #

[Infinately many soln]

$\therefore (n\pi)^2$ is the eigenvalue for the eigunfunction $y(x) = B\sin[(n\pi)x]$

Observing our soln,

$y(x) = B\sin[(n\pi)x]$

$\frac{dy}{dx} = (n\pi)B\cos[(n\pi)x]$

$\frac{d^2y}{dx^2} = -(n\pi)^2 B\sin[(n\pi)x]$

$-\frac{d^2}{dx^2} B\sin[(n\pi)x] = (n\pi)^2 B\sin[(n\pi)x]$

$-\frac{d^2}{dx^2} B\sin[(n\pi)x] = \lambda B\sin[(n\pi)x]$

$\therefore$ When $-\frac{d^2}{dx^2}$ is operated on $B\sin[(n\pi)x]$, we get the scaled up version of the same function, $B\sin[(n\pi)x]$ by $(n\pi)^2$.

Eigenfunction : $B\sin[(n\pi)x]$

Eigenvalue : $(n\pi)^2$

Alternative Method [Recomended]

When $\lambda > 0$

We let $w^2 = \lambda$ ← This is definately > 0, thus we don't lose or create any solns , $w \neq 0$

$\therefore \frac{d^2y}{dx^2} + w^2 y = 0$

Auxiliary Eqn

$r^2 + w^2 = 0$

$r = \pm\sqrt{-w^2}$

$r = \pm wi$

General solution

$y(x) = A\cos(wx) + B\sin(wx)$

Using boundary condition,

$y(0) = A + 0 = 0 \Longrightarrow A = 0$

$y(1) = 0 + B\sin(w) = 0$

$\Longrightarrow B = 0$ or $\sin(w) = 0$

[Reject. Gives trivial soln] same like $\lambda = 0$

$w = n\pi$, where $n \in \mathbb{Z}^+$

Particular solution

$y = B\sin[(n\pi)x]$, for $w = n\pi$

$\Longrightarrow$ Squaring both sides,

We can do this w/out losing any solns bcuz w is guaranteed positive, $w > 0$

$w^2 = (n\pi)^2$

$\lambda = (n\pi)^2$

[Infinately many soln]

$\therefore (n\pi)^2$ is the eignvalue for the eignfunction $y(x) = B\sin[(n\pi)x]$

When $\lambda < 0$

$$\frac{d^2y}{dx^2} - \lambda y = 0$$

Auxiliary Eqn

$$r^2 - \lambda = 0$$

$$r = \pm\sqrt{\lambda}$$

$$= \pm w \text{, where } w = \sqrt{\lambda}$$

General soln

$$y(x) = Ae^{wx} + Be^{-wx}$$

Using boundary conditions,

$$y(0) = A + B = 0 \quad \text{—— ①}$$

$$y(1) = Ae^{w} + Be^{-w} = 0 \quad \text{—— ②}$$

① → ②

$$Ae^{w} - Ae^{-w} = 0$$

$$A[e^{w} - e^{-w}] = 0$$

$$A[2\sinh(w)] = 0$$

$\therefore$ $A = 0$ or $2\sinh(w) = 0$

$\Longrightarrow B = 0$ $\qquad$ $w = 0$

[Reject. Gives trivial soln] same like $\lambda = 0$

[reject $\therefore \lambda < 0$, $\sqrt{\lambda}$ is complex $\Longrightarrow \sqrt{\lambda} \neq 0$, $w \neq 0$]

$\therefore$ There is only 1 e.val $[(n\pi)^2]$ and 1 corresponding e.function $[B\sin(n\pi)x]$ for this ODE.

Alternative Method [Recomended]

$\boxed{\text{When } \lambda < 0}$

we let $-(w^2) = \lambda$, $w \neq 0$

This is definately < 0, sartisfying $\lambda < 0$,

Thus we don't lose generiality

$\therefore \frac{d^2y}{dx^2} - (w^2)y = 0$

Auxiliary Eqn

$r^2 - (w^2) = 0$

$r^2 = w^2$

$r = \sqrt{w}$ or $-\sqrt{w}$

General soln

$y(x) = Ae^{wx} + Be^{-wx}$

Using boundary conditions,

$y(0) = A + B = 0$ —— ①

$y(1) = Ae^{w} + Be^{-w} = 0$ —— ②

① → ②

$Ae^{w} - Ae^{-w} = 0$

$A[e^{w} - e^{-w}] = 0$

$A[2\sinh(w)] = 0$

$\therefore A = 0$ or $2\sinh(w) = 0$

$\Longrightarrow B = 0$ $\quad$ $w = 0$

[Reject. Gives trivial soln] $\quad$ [reject $\therefore w \neq 0$]

same like $\lambda = 0$

Systems of ODE

Linear, 1^{st} order, constant-coefficient ODE systems

$$\left.\begin{aligned}
\frac{dy_1}{dx} &= a_{11}y_1 + a_{12}y_2 + \dots + a_{1n}y_n \\
\frac{dy_2}{dx_2} &= a_{21}y_1 + a_{22}y_2 + \dots + a_{2n}y_n \\
&\vdots \\
\frac{dy_n}{dx} &= a_{n1}y_1 + a_{n2}y_2 + \dots + a_{nn}y_n
\end{aligned}\right\} \quad \frac{d\underset{\sim}{y}}{dt} = A\underset{\sim}{y}, \text{ where } \underset{\sim}{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

If A has eigenvalues $\lambda_1, \lambda_2, \dots \lambda_n$ $\bar{w}$ n linearly independent eigenvectors $\underset{\sim}{v}_1, \underset{\sim}{v}_2, \dots, \underset{\sim}{v}_n$ then the general soln is

$$y = \underbrace{c_1 e^{\lambda_1 x}\underset{\sim}{v}_1 + c_2 e^{\lambda_2 x}\underset{\sim}{v}_2 + \dots + c_n e^{\lambda_n x}\underset{\sim}{v}_n}$$

Arbitary Constants

Why does the solution take this form?

Consider $\frac{dy}{dt} = Ay$

If A has linearly independent eigenvectors, we can diagonalise

$$A = VDV^{-1}$$

matrix whose columns are eigenvectors (V); diagonal matrix of corresponding eigenvalues (D)

$$\Rightarrow \frac{dy}{dt} = VDV^{-1}y \quad \Rightarrow V^{-1}\frac{dy}{dt} = \underbrace{V^{-1}V}_{I}DV^{-1}y$$

$$\Rightarrow V^{-1}\frac{dy}{dt} = DV^{-1}y \quad \Rightarrow \frac{d}{dt}(V^{-1}y) = D(V^{-1}y)$$

write $\underline{x} = V^{-1}y$, then $\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda_1 x_1 \\ \lambda_2 x_2 \\ \vdots \\ \lambda_n x_n \end{pmatrix}$

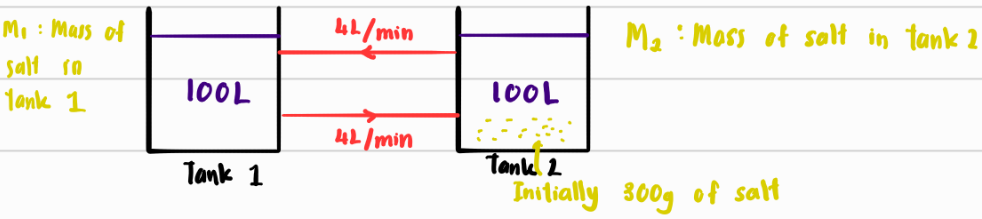
$$\frac{dx_1}{dt} = \lambda_1 x_1 \qquad \frac{dx_2}{dt} = \lambda_2 x_2 \qquad \frac{dx_n}{dt} = \lambda_n x_n$$

$$x_1 = c_1 e^{\lambda_1 t} \qquad x_2 = c_2 e^{\lambda_2 t} \qquad x_n = c_n e^{\lambda_n t}$$

Our original variable $y = V\underline{x} = \left(\underline{v}_1 \middle| \underline{v}_2 \middle| \cdots \middle| \underline{v}_n\right)\begin{pmatrix} c_1 e^{\lambda_1 t} \\ c_2 e^{\lambda_2 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{pmatrix}$

$$= c_1 e^{\lambda_1 t}\underline{v}_1 + c_2 e^{\lambda_2 t}\underline{v}_2 + \dots + c_n e^{\lambda_n t}\underline{v}_n$$

Example 1: [Real distinct e.val]



Since both tank's flow rate is the same, their volume of water will always be 100L.

$$\frac{dM_1}{dt} = -\frac{4}{100}M_1 + \frac{4}{100}M_2$$
$$= -\frac{1}{25}M_1 + \frac{1}{25}M_2$$
$$\frac{dM_2}{dt} = \frac{4}{100}M_1 - \frac{4}{100}M_2$$
$$= \frac{1}{25}M_1 - \frac{1}{25}M_2$$

$$\frac{d}{dt}\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} -1/25 & 1/25 \\ 1/25 & -1/25 \end{pmatrix}\begin{pmatrix} M_1 \\ M_2 \end{pmatrix}$$
$$\frac{d}{dt}\underset{\sim}{M} = A\underset{\sim}{M}$$

Boundary condition : $M_1 = 0$, $M_2 = 300$ at $t = 0$

Eigenvalues :

$$(A - \lambda I) = \begin{pmatrix} -1/25 - \lambda & 1/25 \\ 1/25 & -1/25 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$
$$\left(-\frac{1}{25} - \lambda\right)^2 - \left(\frac{1}{25}\right)^2 = 0$$
$$\vdots$$
$$\Longrightarrow \lambda = 0 \quad \text{or} \quad \lambda = -2/25$$

When $\lambda = 0$

$$(A - \lambda I)\underset{\sim}{M} = \underset{\sim}{0}$$
$$\begin{pmatrix} -1/25 & 1/25 \\ 1/25 & -1/25 \end{pmatrix}\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\Longrightarrow \begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\therefore \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is the corresponding e.vec.

When $\lambda = -2/25$

$$(A - \lambda I)\underset{\sim}{M} = \underset{\sim}{0}$$
$$\begin{pmatrix} -1/25 + 2/25 & 1/25 \\ 1/25 & -1/25 + 2/25 \end{pmatrix}\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 1/25 & 1/25 \\ 1/25 & 1/25 \end{pmatrix}\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

General soln

$$\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = c_1 e^{0t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{(-2/25)t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$
$$= c_1\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{(-2/25)t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\Longrightarrow M_1 = c_1 + c_2 e^{-\frac{2}{25}t}, \quad M_2 = c_1 - c_2 e^{-\frac{2}{25}t}$$

Using boundary conditions

$$0 = c_1 + c_2 e^0, \quad 300 = c_1 - c_2 e^0$$
$$c_1 + c_2 = 0 \text{ --- } ① , \quad c_1 - c_2 = 300 \text{ --- } ②$$

① → ②

$$c_1 = 150$$
$$c_2 = -150$$

$\therefore$ **Particular solution**

$$M_1 = 150 - 150e^{-\frac{2}{25}t} \quad \#$$
$$M_2 = 150 + 150e^{-\frac{2}{25}t} \quad \#$$

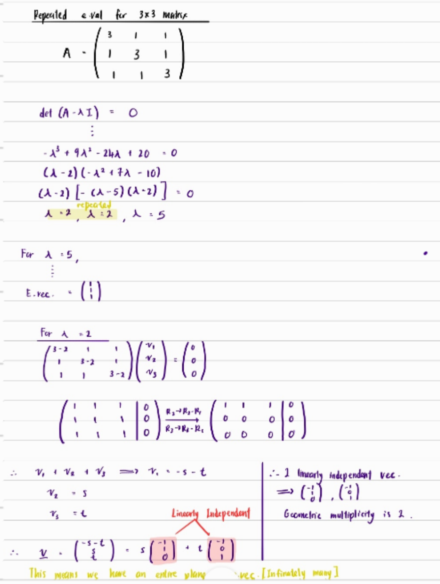
Example 2: [Repeated e.val]

Find the general soln of the system

$$\frac{dx}{dt} = 3x + y + z$$

$$\frac{dy}{dt} = x + 3y + z$$

$$\frac{dz}{dt} = x + y + 3z$$



Repeated e.val for 3x3 matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\vdots$$

$$-\lambda^3 + 9\lambda^2 - 24\lambda + 20 = 0$$

$$(\lambda - 2)(-\lambda^2 + 7\lambda - 10)$$

$$(\lambda - 2)\left[-(\lambda - 5)(\lambda - 2)\right] = 0$$

$$\lambda = 2, \overset{\text{repeated}}{\lambda = 2}, \lambda = 5$$

For $\lambda = 5$,

$\vdots$

E.vec. $= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

For $\lambda = 2$

$$\begin{pmatrix} 3-2 & 1 & 1 \\ 1 & 3-2 & 1 \\ 1 & 1 & 3-2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right) \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\therefore$ $v_1 + v_2 + v_3 \implies v_1 = -s - t$

$v_2 = s$

$v_3 = t$

Linearly Independent

$\therefore$ $\underline{v} = \begin{pmatrix} -s-t \\ s \\ t \end{pmatrix} = s\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

This means we have an entire plane [illegible] vec. [Infinitely many]

$\therefore$ 2 linearly independent vec.

$\implies \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

Geometric multiplicity is 2.

Matrix form

$$\frac{d}{dt}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\frac{d}{dt}\underset{\sim}{x} = A\underset{\sim}{x}$$

After some calculation, eval & e.vec of A is

$\lambda_1 = 2$, $\underset{\sim}{v_1} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$

$\lambda_2 = 2$, $\underset{\sim}{v_2} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

$\lambda_3 = 5$, $\underset{\sim}{v_3} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

General soln : $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 e^{2t}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{2t}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + c_3 e^{5t}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ ✗

Euler's Formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Example 3 : [Complex e.val]

Solve $\dfrac{dx}{dt} = x + y$

$\dfrac{dy}{dt} = -4x + y$

Subject to $x = 1$, $y = 1$ at $t = 0$

Matrix Form

$$\frac{d}{dx}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -4 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{d}{dx}\underset{\sim}{x} = A\underset{\sim}{x}$$

After calculation, the e.vec & e.val of A is

$\lambda_1 = 1 + 2i$, $\underset{\sim}{v_1} = \begin{pmatrix} 1 \\ 2i \end{pmatrix}$

$\lambda_2 = 1 - 2i$, $\underset{\sim}{v_2} = \begin{pmatrix} 1 \\ -2i \end{pmatrix}$

General soln

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^{(1+2i)t}\begin{pmatrix} 1 \\ 2i \end{pmatrix} + c_2 e^{(1-2i)t}\begin{pmatrix} 1 \\ -2i \end{pmatrix}$$

Using boundary conditions

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = c_1\begin{pmatrix} 1 \\ 2i \end{pmatrix} + c_2\begin{pmatrix} 1 \\ -2i \end{pmatrix}$$

$\Longrightarrow c_1 + c_2 = 1$, $(2i)c_1 - 2i\,c_2 = 1$

$\vdots$

$c_1 = \frac{1}{2} - \frac{1}{4}i$, $c_2 = \frac{1}{2} + \frac{1}{4}i$

Particular soln

$$\begin{pmatrix} x \\ y \end{pmatrix} = \left(\tfrac{1}{2} - \tfrac{1}{4}i\right)e^{(1+2i)t}\begin{pmatrix} 1 \\ 2i \end{pmatrix} + \left(\tfrac{1}{2} + \tfrac{1}{4}i\right)e^{(1-2i)t}\begin{pmatrix} 1 \\ -2i \end{pmatrix}$$

$$= \left(\tfrac{1}{2} - \tfrac{1}{4}i\right)e^{t}\overset{\cos(2t) + i\sin(2t)}{\left(e^{2ti}\right)}\begin{pmatrix} 1 \\ 2i \end{pmatrix} + \left(\tfrac{1}{2} + \tfrac{1}{4}i\right)e^{t}\overset{\cos(2t) - i\sin(2t)}{e^{(-2ti)}}\begin{pmatrix} 1 \\ -2i \end{pmatrix}$$

$\vdots$

$x(t) = e^{t}\left(\cos(2t) + \frac{1}{2}\sin(2t)\right)$ #

$y(t) = 2ie^{t}\left[i\sin(2t) - \frac{1}{2}i\cos(2t)\right]$

$y(t) = e^{t}\left[-2\sin(2t) + \cos(2t)\right]$ #